Zhang et al. Large-scale Assessments in Education (2026) 14:19
https://doi.org/10.1186/s40536-026-00287-w

RESEARCH

Open Access

Evaluating generative AI chatbots for large-scale assessment data: comparing LLM-as-a-judge and human ratings

Ting Zhang[1*], Luke Patterson[2], Blue Webb[2], Zeyu Jin[2] and Maggie Beiting-Parrish[3]

*Correspondence:
Ting Zhang
ting.zhangumd@gmail.com
[1]College Board, New York, United States
[2]American Institutes for Research, Washington, DC, United States
[3]Federation of American Scientists, Washington, DC, USA

Abstract

Introduction This study focuses on developing and evaluating a customized Generative AI chatbot designed to enhance access to large-scale educational data. The chatbot aims to assist researchers and policymakers in exploring complex datasets, such as NAEP, through natural language queries.

Methods The chatbot was built using a Retrieval-Augmented Generation (RAG) framework that integrates multiple specialized agents to retrieve, interpret, and synthesize educational data. One agent was selected as a case study for performance evaluation. The study compared an automated Large Language Model (LLM)-based evaluation ("LLM-as-a-judge") with human expert ratings to examine validity and consistency across three criteria: correctness, completeness, and communication quality. A total of 141 expert-generated questions reflecting typical user queries were used, each accompanied by a reference answer and source documentation. Chatbot's responses were evaluated with a three-dimensional framework on Correctness, Completeness, and Communication. In addition to human evaluation, an LLM-based evaluation was implemented, and the model was provided with the rubric, human-written reference answers, and retrieved RAG contents to generate automated quality assessments. Interrater reliability among human raters and the LLM-as-a-judge were computed with quadratic weighted kappa (QWK).

Results Findings showed that the LLM-as-a-judge approach achieved comparable agreement levels with human raters and demonstrated reliability across all evaluation dimensions. Interrater reliability analyses revealed no significant differences between inter-human and human-to-LLM agreement, except in the communication dimension, where human-to-LLM consistency was higher. These results indicate that the LLM-as-a-judge method can serve as a viable and consistent alternative to human evaluation for customized RAG-based chatbot assessment.

Conclusions Integrating LLM-based evaluation into the assessment of Generative AI chatbots provides a scalable, reliable, and cost-effective complement to traditional human review. With human oversight for calibration and validation, this approach enables more efficient and consistent evaluation practices, advancing the use of AI tools that facilitate broader access to large-scale educational data.

Keywords Artificial intelligence, Retrieval-augmented generation, Human rater, Automated evaluation, LLM-as-a-judge, Interrater reliability, QWK

Introduction

Each year, national and international education agencies produce a wealth of valuable data and statistical information. However, accessing specific information remains a challenge for many users. Individuals often struggle to find clear answers to seemingly simple questions like "How do NAEP assessments benefit my child's school and community?", "Are the same districts selected by NAEP from year to year?", or "Which school districts in the U.S. experienced the most significant decline in 2024 math scores?" The answers to these questions are frequently scattered across multiple platforms, documents, and formats, such as narrative reports, tables, figures and data files, making them difficult to locate and synthesize. This creates a critical need for more user-friendly tools such as Generative Artificial Intelligence (Generative AI) chatbots that can improve access to educational data, especially for users who may not be familiar with technical language or where to look.

Generative AI tools such as chatbots offer a promising solution by enabling more efficient and intuitive access to information. However, standard large language models (LLMs) have limitations. They can generate inaccurate or misleading responses and are restricted by the static nature of their training data. These weaknesses are especially problematic in environments where accuracy, objectivity, and timeliness are paramount.

For agencies and institutions that use education data to compare student achievement, evaluate program effectiveness, or provide educational resources, ensuring the Gen AI tools access to accurate and up-to-date information is essential. Moreover, because these agencies are tasked with reporting data in a manner that is objective, neutral, and free from political bias, it is crucial that any tools used to assist users uphold these standards.

To address these challenges, retrieval-augmented generation (RAG) models offer a powerful alternative to foundational LLMs (Vach, et al., 2025; Zhang et al., 2024). RAG enhances generative AI by grounding responses in authoritative, up-to-date external sources, such as federal databases or verified documentation. This enables the chatbot to understand natural language questions and provide precise, context-aware answers drawn from verified data rather than relying solely on the LLM's internal knowledge.

AIR developed Ask NCES, a multi-agent chatbot that supports a broad range of national and international education assessments. The evaluation of one agent, Ask NAEP, has demonstrated that the agent significantly improves response quality, helping users navigate NCES data and documentation more effectively by providing accurate answers with relevant sources (Zhang et al., 2024). However, a key lesson from this research is that human-based evaluation is both costly and time-intensive. In our prior work, the effort required to evaluate Ask NAEP was approximately two to three times greater than the effort devoted to its development. As we extend this chatbot to a multi-agent paradigm, such an evaluation approach becomes increasingly impractical under existing resource constraints. This necessitates the exploration of more efficient evaluation methods that do not compromise evaluation quality. In this context, the LLM-as-a-judge approach represents a promising alternative worthy of systematic investigation.

Purpose and scope

This study details how we applied the Ask NAEP approach (Zhang et al., 2024) to develop another Generative AI agent, Ask CSO, designed to answer questions related to the NCES Statistical Standards Program overseen by the Chief Statistician's Office (CSO).

Leveraging the evaluation framework established for Ask NAEP, this study investigates a hybrid evaluation approach incorporating both human expert assessments and LLM-based automated evaluations. The objective was to assess the quality of Ask CSO's responses and determine if an LLM-driven evaluation process could enhance efficiency without compromising the quality of chatbot performance assessments.

The Statistical Standards Program, led by the Chief Statistician's Office (CSO), ensures the quality, integrity, and confidentiality of NCES statistical activities through the development of standards, expert review, and oversight of data access. Most the program materials are available online. (National Center for Education Statistics, n.d.), but presented in various formats, including PDFs, HTML texts, and tables. Different versions, such as the 2002 (the outdated version) and 2012 NCES Statistical Standards, are also available, making it difficult to locate and retrieve the most relevant and updated information.

Research questions

We tested the following research questions (RQs) in this study:

RQ 1. Does the Ask CSO retrieval process and bot configuration produce quality answers? We first want to establish that Ask CSO successfully produces quality answers, as evaluated by expert human reviewers based on our evaluation framework and rubric.

RQ 2. Are there any automated text evaluation metrics that are strong proxies of human assessment of Ask CSO answers? Human assessment of all question/answer pairs is labor intensive, so we sought automated evaluation metrics that could be strong proxies for human assessment. Automated evaluation metrics are metrics of answer text quality that can be calculated without human review of a given question/answer pair.

RQ 3. How does the quality of Ask CSO answers compare with that of a non-retrieval-augmented GPT-4o model? It may be argued that the only component that needs to be evaluated properly is the generative component and how effective the generated responses are. In a RAG-powered chatbot, however, the retrieval is an important intermediary that can help diagnose why a chatbot responds correctly or incorrectly to queries. If receiving the correct retrieval is unimportant, RAG is not providing a significant improvement over unaltered GPT models, so testing the retrieval is one of the evaluation's research questions.

RQ 4. Does adding examples and chain-of-thought to the rubric improve the LLM evaluator? Previous literature has suggested that LLM applications produce better quality results when supplied with examples and the reasoning behind those examples. We tested what happened when we incorporated such changes to our rubric prompts for the LLM evaluator.

RQ 5. How consistent is the LLM evaluator across successive iterations with identical parameters? LLM text generation has some stochastic components, and will produce different results across successive iterations even with identical parameters. We tested the consistency of LLM evaluations across multiple iterations to see if scores varied significantly.

Development process

Architecture

Ask CSO is a retrieval augmented generation (RAG) chatbot. In RAG, user queries are not sent directly to a generative model for a response. Rather, queries are used to retrieve relevant and verified documents from a vector database called knowledge base using a search algorithm, and the retrieved documents alongside the query are provided to a generative model. Foundational generative models, such as the web browser based GPT models, are by nature generalists. The large scale of data that GPT and similar models have been trained on makes them well suited for answering high level queries about a wide array of topics. However, they are less effective for detailed, domain-specific questions and are known to occasionally produce inaccurate answers and hallucinations. In contrast, RAG models can provide more trustworthy answers by retrieving information from a curated database, thereby reducing the hallucination. RAG has become a popular choice for reaping the benefits of context awareness and the advanced language understanding inherent in pre-trained models.

In a RAG chatbot workflow, the user submits a query through the interface, which is then preprocessed for optimal retrieval and generation. The retrieval module searches an external knowledge base to gather relevant information, which is combined with the original query to enhance context. This augmented input is processed by a generative model, such as an open AI GPT-4o, to produce a coherent and contextually relevant response. Finally, the generated response is delivered back to the user, completing the interaction.

We advanced the basic RAG framework by including a knowledge base containing publicly available, verified information from NCES websites, and a guardrail for filtering queries based on relevancy. Queries relevant to NCES are accepted and used for retrieval, while queries deemed irrelevant are rejected and a preset response is returned to the user. This ensures that resources are not needlessly consumed, as irrelevant queries will not result in API calls to the text generation endpoint. Figure 1 shows the overall architecture of Ask CSO.

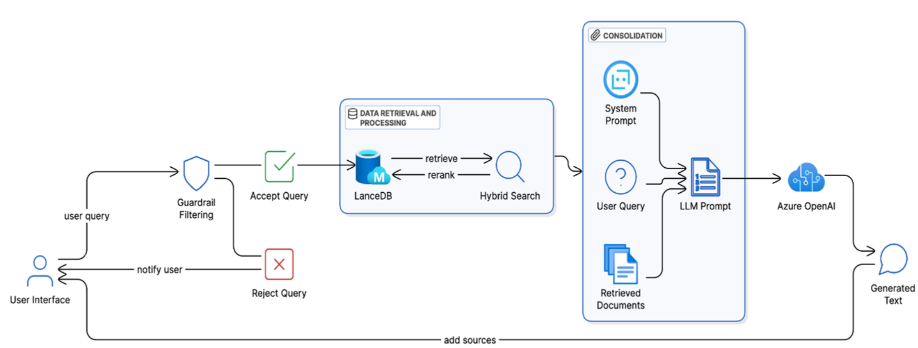

Fig. 1 Ask CSO Architecture

Appendix A provides additional details on the chatbot's development, including its user interface design, frontend and backend architecture, and the measures implemented to ensure the security of its knowledge base.

Data collection

Leveraging the *requests* and *BeautifulSoup* modules in Python, we web crawled the *Statistical Standards Program* section of the NCES site to obtain a list of URLs for both web pages and PDFs. This amounted to ~ 340 web pages and 65 PDFs. Each web page and PDF was then processed to extract text and other relevant information. We developed a custom pre-processing pipeline for extracting the primary text content from webpages, which handles the removal of any unneeded page elements (e.g. navigation bars, side panels, footers) and extracts table elements for further processing. These pre-processed texts were saved in a readable format using the *html2text* Python module. Any tabular data extracted was parsed into a JSON format and stored as metadata for the associated page. Alt text for figures was likewise captured. The page URL and title were retained as additional metadata.

For PDFs, all body text and table text were extracted as a single entity using the state-of-the-art *PyMuPDF* Python module. Table data from PDFs was not parsed into a structured format due to the inconsistency in how table structures are embedded in PDFs, but the robust extraction of table text ensures semantically meaningful embeddings. Metadata captured for PDFs consisted of the PDF URL and page number.

Evaluation methodology

Overview and literature review

The evaluation of LLMs is a multifaceted area with ongoing research and development. Traditional evaluation methods include N-gram Matching, which involves counting the occurrences of n-grams (sequences of n words) in the generated text and comparing them to the reference text (Cavnar & Trenkle, 1994). N-gram matching metrics like BLEU (Bilingual Evaluation Understudy, Papineni et al., 2002) and ROUGE (Recall-Oriented Understudy for Gisting Evaluation, Lin, 2004) are well-established and widely used in the field of natural language processing (NLP). However, the method has limitations in understanding semantics and context, making it less effective for evaluating more complex or nuanced language generation tasks (Barbella & Tortora, 2022; Reiter, 2018). For example, it does not consider the broader context or coherence of the text, which is crucial for evaluating the quality of generated responses in conversational AI. It also can be overly rigid, failing to capture the fluidity and creativity of natural language. It may not adequately evaluate texts that are correct but use different phrasing.

Human evaluation remains the gold standard for assessing the quality of generated texts. It involves human reviewers rating responses based on criteria such as correctness, completeness, coherence, and fluency (Celikyilmaz et al., 2021; Howcroft & Rieser, 2021; Iskender et al., 2021; Schoch et al., 2020; Smith et al., 2022; van der Lee et al., 2021; Zhang et al., 2024). A well-trained human expert can assess the relevance and appropriateness of responses within specific contexts. However, human evaluations can be subjective, leading to variability in ratings across different evaluators. Furthermore, conducting human evaluations is time-consuming and requires significant effort, especially for large datasets.

Recent research has explored using LLMs themselves to evaluate generated texts, known as the LLM-as-a-judge framework (Li et al., 2024, Szymanski et al., 2025). This approach leverages the capabilities of LLMs to mimic human reasoning and decision-making processes for evaluation purposes. This approach has the promise of offering consistent scoring and can be the substitute human evaluator for processing large volumes of data. But limited studies have been done in this area to compare its quality with human evaluation. Additionally, cutting-edge approaches to using LLMs in evaluation of generated text sometimes use multi-agent "teams" of agents that evaluate and give responses to text based on different "personas" or viewpoints (Chan et al., 2023). This is seen as comparable to the human evaluation process of using two or more raters to ensure that no one perspective is adding too much subjectivity into the output (Liang et al., 2023). It is also a potential substitute for needing large teams of human raters, if the different evaluator personas can fulfill some of the traditional roles a human rate would.

Furthermore, LLM evaluation of output can be improved using a few annotated examples to support LLM "understanding" of criteria at different levels of output quality. For example, with a "few shot" approach, the system is provided with some annotated examples of what makes for low vs. high quality examples and how these should be annotated (Fu et al., 2023); this helps improve the ability of the LLM(s) for evaluation because they have some concrete examples of what quality output looks like for each specific scoring band.

The Chain-of-Thought (CoT) prompting is another technique for improving the evaluation of LLM-generated output. In this approach, the LLM evaluator is prompted to explain the reasoning behind the evaluation(s) given in a step-by-step approach. This can be used to generate more accurate scores that match human evaluation. The results of CoT prompting can be close to that of human evaluators but the actual "reasoning" that is described may be partially or largely incorrect (Ngyuen et al., 2024).

While there are many different approaches to using LLMs as evaluators, no one process has emerged as a "best practice", and it is good to use several approaches.

Validation techniques and metrics

To address our research questions listed in Sect. 1.3, we built a test set of 141 queries (noted as $\sum_{i=1}^{141} Question_i$). These questions were developed by content experts across five topic areas covered by the NCES Statistical Standards Program. The experts had supported the CSO in multiple capacities, including responding to inquiries from users seeking guidance on NCES assessments and survey-related questions. They were instructed to formulate queries from the perspective of typical CSO website users. Table 1 lists examples of queries broken up into the five topics.

These experts also provided a narrative answer (referred as the ground truth answer) as well as its source links (annotated as $A_{i,e} \forall i \in \{1, \ldots, 141\}$, where A indicates it is an answer, the *i* indexes the question, the subscript *e* indicates it is a human expert answer) to each question. We fed the test questions to the latest version of the Ask CSO chatbot (Bot_{pub}), and several experimental variations of the chatbot.

Evaluation metrics

Ideally, communicating with a chatbot should feel like a natural conversation, in which the chatbot's written product is as comprehensible as a text message communication

Table 1 Example test queries

Topic	Example question
Overview	What does the Statistical Standards Program provide? How does the Program develop standards? What are the responsibilities of the Senior Official?
Statistical standards	When were the Written Standards first developed? At each stage of the data editing process, what kinds of data checks are needed? When is a nonresponse bias analysis required? What are some examples of machine-readable products? When do I need to present the key variables? What must the initial plan for developing a survey or survey system include? At each stage of the data editing process, what kinds of data checks are needed? What are all of the elements needed for the analysis plan?
IES style guide	When should I spell out acronyms in my report? How should I correctly use the terms "percent" and "percentage" in my report?
Contents of NCES surveys	What are the four skill domains on the ALL?
Restricted-use data licenses	What counts as "restricted-use" data? Who are the required personnel to apply for an RUD license? If I need an amendment to a current license, which system should I use? Is there a review process for draft documents containing restricted use data? What counts as a qualified organization in the Restricted-Use Dataset process? What are the requirements for the "project office" in an RUD application? What are the expectations of every person who has access to the RUD?

Table 2 Framework for generative component evaluation

Construct	Annotation	Description
Correctness	$Q_{correct}$	Is the content of the chatbot's answer factually correct?
Completeness	$Q_{complete}$	Does the chatbot's answer contain relevant facts and information to answer the question?
Communication	Q_{comm}	Is the chatbot's answer written in a clear and concise fashion?

produced by a human author. In thinking about this, we created an initial framework that used Grice's Maxims of Conversation (1989); this framework sees a conversation as a shared product between both parties who have some common aim in mind. In this case, the common goal is to better understand some aspects of CSO content, whether it is more procedural information or a question around specific test results. Within this conversational exchange, there are four maxims that ensure a quality response: quantity, quality, relation, and manner (Grice, 1989) These are especially relevant to the presentation of statistical chatbot responses because these should ideally be long enough to include all of the necessary information without being burdensome (quantity), truthful (quality), only include relevant information (relation), and be as concise and clear as possible (manner). Several of these criteria are very specific to the individual user; so, this system was simplified to three criteria: Correctness, Completeness, and Communication, as outlined in Table 2 below.

The following weights are used to generate a composite score from these three constructs, that place more weight on correctness and completeness than on communication to mirror the federal agency NCES's emphasis on the importance that the chatbot provide correct and complete answers (Zhang et al., 2024).

$$Q_{composite} = \frac{2}{5}Q_{correct} + \frac{2}{5}Q_{complete} + \frac{1}{5}Q_{comm}$$

This approach is consistent with measurement frameworks in which primary validity threats (incorrect or incomplete content) are weighted more heavily than secondary

Table 3 Dimension scoring rubric

Grade	Correctness	Completeness	Communication
1 (Poor)	Significant factual errors or misinformation	Incomplete, missing crucial information	Unclear, convoluted, or difficult to follow
2 (Below Average)	Some inaccuracies or lack of precision	Lacks relevant details or fails to address all aspects	Lacks coherence and may confuse the reader
3 (Average)	Several minor inaccuracies, but draws the generally correct conclusion	Covers basics but could benefit from more depth	Clear but could be more concise
4 (Good)	Generally accurate with minor inaccuracies	Sufficiently complete, addressing main points	Generally clear and concise
5 (Excellent)	Completely correct with no errors	Comprehensive with thorough information	Succinct, well-organized, and easy to understand

Table 4 Pass rates (grade of > = 3)–human assessment of ask CSO quality

Dimension	Pass	Fail	Total	Pass %
Correctness	170	28	198	86
Completeness	169	29	198	85
Communication	178	20	198	90
Overall	170	28	198	86

Total column reflects all human reviews given to all questions (excluding a few where the reviewer was not able to make a determination). Multiple humans reviewed some of the questions, which explains the higher number of reviews compared to the number of questions

quality dimensions such as clarity or style (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 2014; Cook & Beckman, 2006).

Three human raters provided an assessment for each of the 141 Bot_{pub} answers on each of the dimensions on a numeric rating scale of 1 to 5, using the expert-provided narrative answer and links as context. The rubric used for this scale is as follows in Table 3.

$$Human_{i,Botpub,q} \in [1{,}5]\,,\ where\ i \in [1{,}141]\ and\ q \in [Q_{correct}, Q_{complete}, Q_{comm}]$$

We also created the "pass/fail" grading for some evaluation analyses. The rubric was aggregated so that grades of 3, 4, and 5 represent qualitatively acceptable answers for a published chatbot, while grades of 1 and 2 do not. This is why the threshold for a passing answer is 3 or higher for all dimensions.

Interrater reliability was assessed using quadratic weighted kappa (QWK), selected for its suitability for ordinal rating scales and its ability to account for chance agreement while weighting the magnitude of disagreement (Table 4).

Human evaluation

Three human raters evaluated each of the 141 public chatbot responses using a 5-point ordinal scale (1 = lowest, 5 = highest). Ratings were provided for correctness, completeness, and communication quality, using expert-authored narrative answers and reference links as evaluation benchmarks:

$$Human_{i,Botpub,q} \in [1{,}5]\,,\ where\ i \in [1{,}141]\ and\ q \in [Q_{correct}, Q_{complete}, Q_{comm}]$$

Prior to formal evaluation, raters were instructed to review the evaluation rubric and practice scoring independently. Special attention was given to handling ambiguous cases, with raters asked to prioritize the reference answers and links. When uncertainty remained, they were instructed to apply the rubric conservatively and documented their rationale for discussion.

Calibration was conducted using 30 selected queries across the five topics, which were independently rated by all three raters. Calibration meetings were held by the project director with the raters to review discrepancies, clarify rubric interpretation, and refine shared understanding of rating standards.

Final interrater reliability statistics among human raters are reported in Table 5. Following calibration, the remaining items were divided into three subsets, with each rater independently evaluating one subset. Throughout the evaluation, raters worked independently and were blind to model identity to minimize potential bias.

Automated evaluation methods

For automated evaluation, we used an LLM-as-a-judge method. The evaluation package Ragas (Es et al., 2023) offers an LLM-as-a-judge method based on a rubric (i.e., rubrics-based criteria scoring) with a metric used to evaluate responses on a specific domain. Chatbot responses are scored using an LLM which is given the rubric, the ground truth answer, and the context retrieved. In our application, we provide the rubric described Table 3 above, the answer written by subject experts, and the content the RAG retrieval process retrieves, respectively. Ragas evaluator LLM objects are initialized with default parameters. Ragas' default LLM decoding parameters are selected to make responses close to deterministic; temperature is set very low (.01). "top_p" is set to .1, ensuring further token selection is done conservatively.

We used the following prompt in Ragas to instruct the LLM to assign a grade:

"Given user input, response and reference that's desired outcome that gets a score of 5, and a score rubric representing evaluation criteria are given.

1. Write detailed feedback that assesses the quality of the response strictly based on the given score rubric, without evaluating in general.
2. After writing the feedback, assign a score between 1 and 5, referring to the score rubric."

Due to resource constraints, Research Question 3 (RQ3) used only automated evaluation; no human evaluation was conducted for this research question.

Results

RQ1. Does the Ask CSO retrieval process and bot configuration produce quality answers? Ask CSO attempted to answer all of the test questions, and human raters gave generally high rating scores to these answers across all dimensions. Table 4 presents these results, and Fig. 2 provides a histogram showing the frequency of each grade for every dimension. Ask CSO achieved a pass rate of 86% in correctness, indicating that the majority of its answers were accurate and met the expected standards. 85% of Ask CSO's

Table 5 Interrater agreement between humans and LLM reviewers

Dimension	Inter-Human QWK	Human-to-LLM QWK
Correctness	0.62	0.69
Completeness	0.66	0.61
Communication	0.45	**0.66
Overall	0.59	0.64

Difference is significant at the **5% confidence level

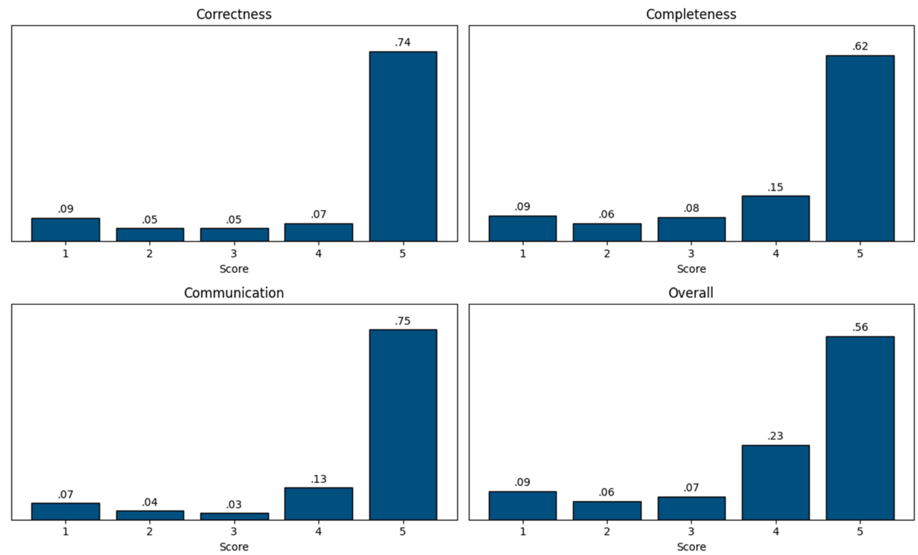

Fig. 2 Mass density functions—human assessment of ask CSO quality

answers received passing grades for the completeness dimension. The communication dimension received the highest pass rate, at 90%. The overall pass rate of 86% reflects a strong performance across all dimensions, suggesting that Ask CSO is a reliable tool for answering questions effectively.

RQ2. Are there any automated evaluation metrics that are strong proxies of human assessment of Ask CSO answers? Because any two human raters will differ in ratings, we do not expect automated ratings to achieve perfect correlation with human ratings. A second human also reviewed each question response. We then measured interrater reliability (IRR) with quadratic weighted kappa (QWK). The QWK was conducted between the two human reviewers (inter-human) and it was compared to the QWK of each human to the LLM reviewer (human-to-LLM). To test the statistical significance of the differences, we used bootstrap sampling with 10,000 iterations to create a distribution of kappa values, calculated the standard deviation of the bootstrap samples, and compared the kappa values with t tests. The human-to-LLM IRR was not significantly different at the 95% level from inter-human IRR for all dimensions except for communication, where the human-to-LLM reviewers had significantly higher interrater agreement. Figure 3 compares the mass density of human and automated scores for all dimensions.

RQ3. How does the quality of Ask CSO answers compare with the quality of answers from GPT-4o? After we established the credibility of the LLM-as-a-judge approach, we expanded our evaluation to compare Ask CSO with a baseline GPT-4o model configured with a system prompt describing it as a helpful assistant for answering questions about NCES CSO policy. In this phase, we examined whether Ask CSO outperforms the baseline model. As a reminder, the dimension scores shown in this section are the LLM-assessed scores, and a passing score is a 3, 4, or 5 for the dimension. In this evaluation round, we applied the Ragas LLM-as-a-judge automated evaluation method for comparisons.
Table 6 shows that the Ask CSO retrieval process led to statistically significant improvements in passing answer percentages on all dimensions. Figure 4 compares the mass density of Ask CSO and GPT-4o scores for all dimensions.

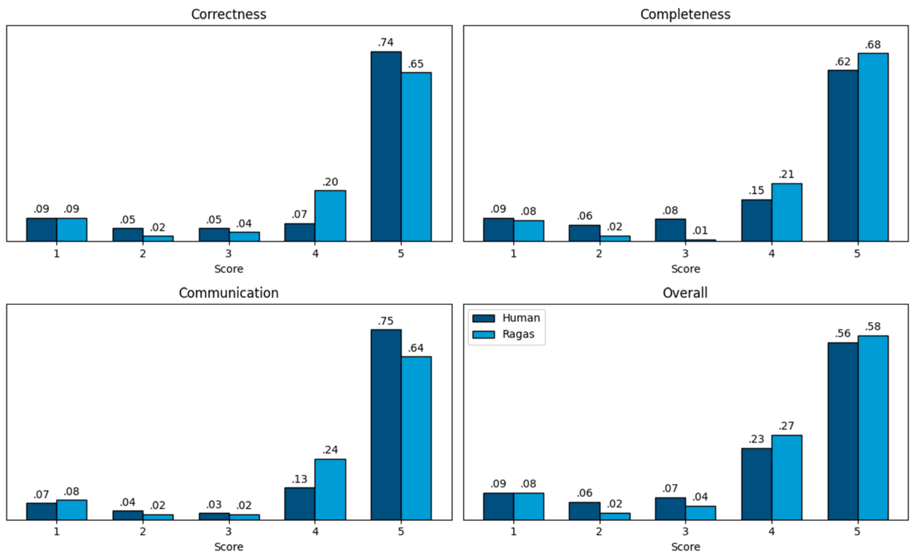

Fig. 3 Mass density functions—comparing human and LLM (Ragas) evolution of ask CSO quality

Table 6 Comparing pass rates for GPT-4o and ask CSO

	N	GPT-4o (%)	AskCSO (%)	p-value	Difference (%)
Correctness	141	77	89	0.003	+12
Completeness	141	82	90	0.032	+8
Communication	141	82	90	0.046	+8
Overall	141	76	89	0.002	+13

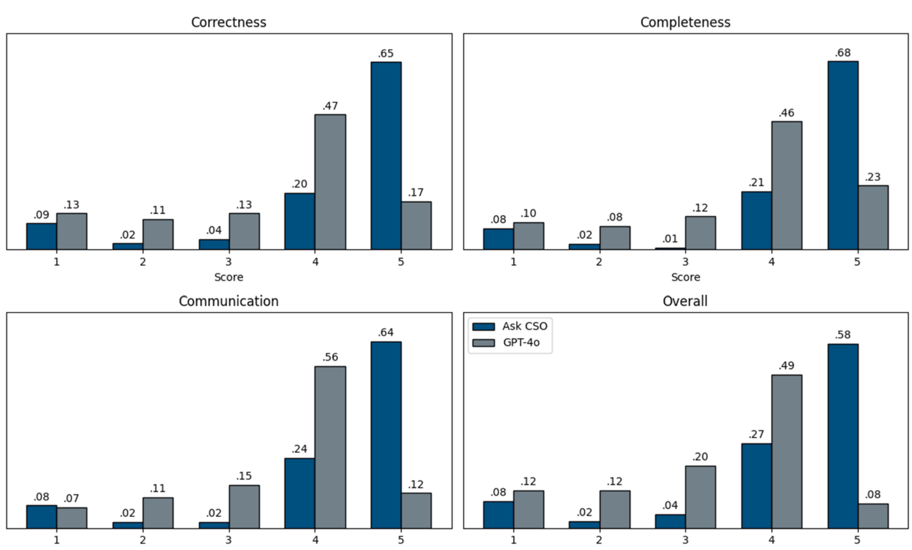

Fig. 4 Mass density functions—comparing GPT-4o and ask CSO quality scores

RQ 4. Does adding examples and chain-of-thought to the rubric improve the LLM evaluator? In this research question, we tested the effects of different rubric types given to the LLM evaluator (Ragas) on the human-to-LLM interrater reliability. We tested three different types of rubrics (see Appendix B for detailed examples of the different types):

1. "No Examples" rubric–just rubric with no examples

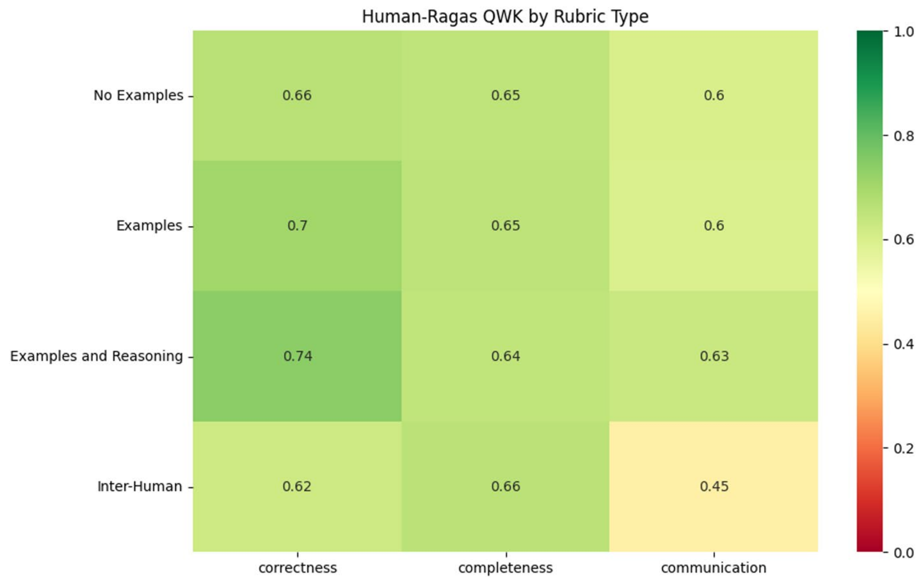

Fig. 5 Human-LLM QWK by rubric type

Table 7 Bootstrap resampling p-values

	Correctness	Completeness	Communication
No examples	0.099*	0.84	0.002***
Examples	0.001***	0.79	0.01**
Examples and reasoning	0.001***	0.84	0.016**

Difference is significant at the *10% confidence level, **5% confidence level, *1% confidence level

2. "Examples" rubricadd 3 example question/answer pairs along with rubric criteria, with human-assigned evaluation score.
3. "Examples and Reasoning" rubric–same as "Examples", but humans also explain their thought process for assigning the score.

Using the 141 Ask CSO test questions and human ratings, we ran Ragas automated evaluation on the test question set 3 times, using a different rubric type each time. We then compared human-LLM and inter-human IRR results, as well as IRR among rubric types, as shown in Fig. 5.

Figure 5 shows human-to-LLM QWK for each of the three rubric types, as well as inter-human QWK for comparison. For correctness, we see slightly higher QWK values for all three LLM rubrics, with slight increases as the rubrics get more complex (No rubrics being the least complex and Examples and Reasoning being the most complex). Completeness QWK is about the same for all rubric types and are similar to inter-human QWK. All three rubrics have significantly higher QWK than inter-human QWK for the Communication dimension. The statistical significance of these results is shown in Table 7, using a one-sided bootstrap resampling significance test. The null hypothesis of this test is:

$$RubricHumanQWK_{t,dim} - InterHumanQWK_{dim} \leq 0$$

where t is the type of rubric and dim is the dimension. This test is repeated for each combination of rubric type and dimension.

Stronger interrater reliability between LLM and humans than IRR among humans could mean one of a few different things. The LLM evaluator might be more consistent in its evaluations compared to human raters, who can be influenced by subjective factors and individual biases. Automated methods like LLM-as-a-judge can apply evaluation criteria uniformly across all responses, whereas human graders might interpret and apply criteria differently. LLM evaluators like Ragas might be designed to focus on specific, measurable aspects of chatbot output quality, leading to more precise and repeatable assessments. However, human raters can understand nuances, context, and subtleties that an automated system might miss. High IRR does not necessarily mean better overall evaluation quality. Further research may be warranted to investigate the underlying reasons for higher IRR.

We also investigated IRR among the different LLM rubric evaluators as well. Figure 6 shows the IRR among all pairs of different evaluators for all three dimensions. Overall, these results show extremely high correlation among the different rubric types. The lowest QWK among any pair of evaluators on any dimension was .86, which is still very strong agreement. This demonstrates that all rubric types gave similar evaluation results on the sample of Ask CSO test questions. In practice, this implies that rubric complexity should be chosen based on a cost–benefit trade-off, with simpler rubrics favored for scalable deployment and more complex rubrics reserved for cases where small gains in correctness or interpretability are critical. The high inter-rubric agreement suggests that simpler rubrics are sufficient to achieve strong human–LLM alignment for this task, with more complex rubrics yielding only marginal improvements (primarily for correctness) at substantially higher annotation cost.

RQ 5. How consistent is the LLM evaluator across successive iterations with identical parameters? To answer this question, we ran the Ragas evaluator with "No Examples" rubric over all 141 Ask CSO test questions ten times, with no changes in parameters. We then measured the QWK of each dimension across all ten iterations. As QWK is a pairwise metric, the overall QWK was computed by calculating the QWK of all possible combinations of pairs of the ten iterations, then taking the mean result of all pairs. The results are shown in Table 8. The QWK values are all extremely high, with no dimension scoring lower than 0.97. This indicates that there is very little variance due to purely stochastic factors in the LLM evaluation for Ask CSO questions.

Discussion

Summary and evaluation results

This study outlines the architecture, technology, and evaluation of Ask CSO, a sub-agent within Ask NCES. Ask NCES is a Generative AI-powered chatbot that utilizes the RAG framework to retrieve and synthesize publicly available data and information from the NCES. As an extension of Ask NAEP (Zhang et al., 2024), the first sub-chatbot developed within Ask NCES, Ask CSO is specifically designed to answer queries related to the NCES Statistical Standards Program using publicly available information from its website (https://nces.ed.gov/statprog/).

We evaluated Ask CSO using the Ask NAEP framework (Zhang et al., 2024), which assesses three dimensions: Correctness, Completeness, and Communication. Using a rubric based on this framework, three human raters scored chatbot responses on an

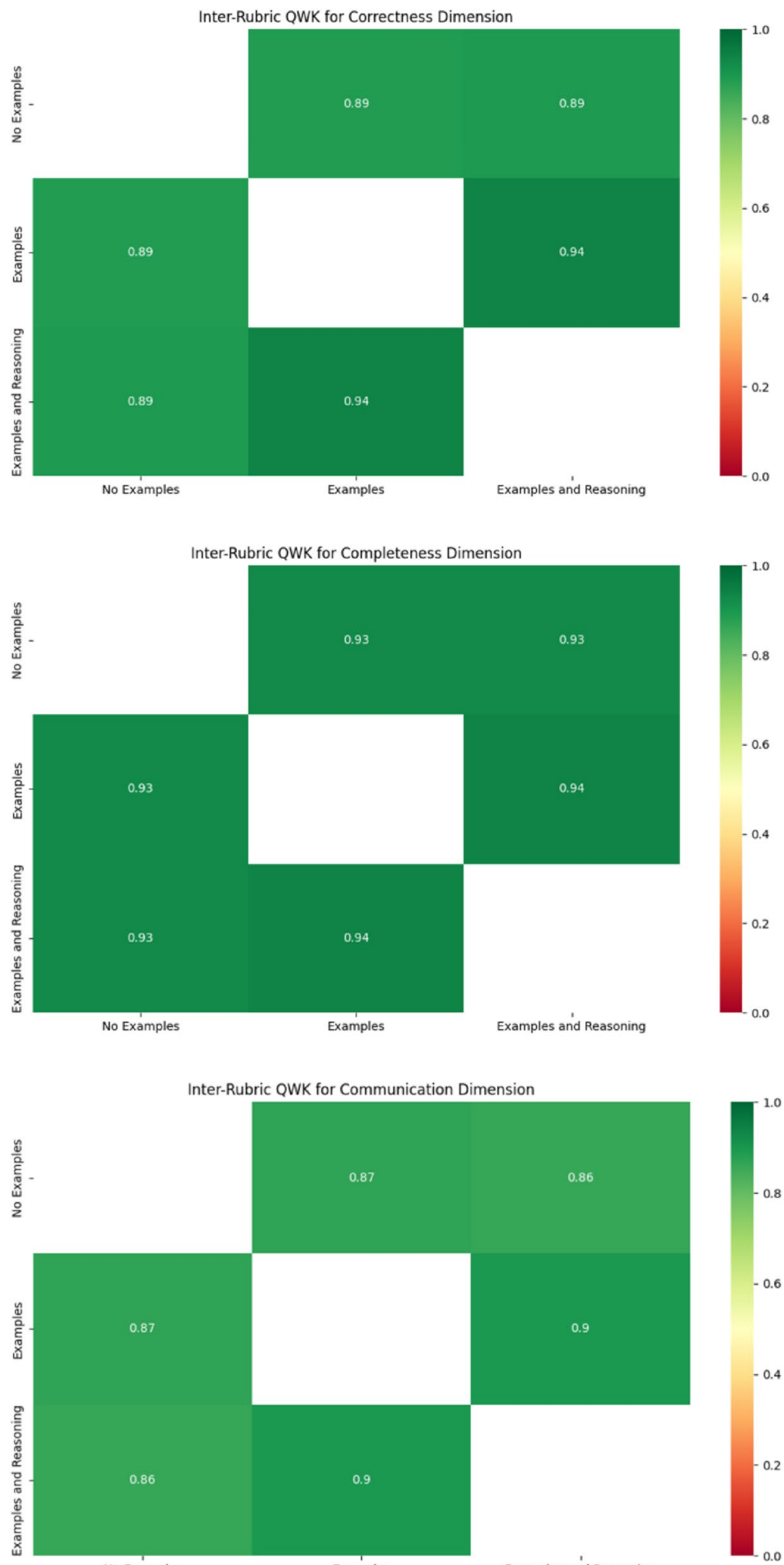

Fig. 6 Inter-Rubric QWK by dimension

Table 8 Average inter-iteration QWK by dimension

	N	QWK
Correctness	141	0.97
Completeness	141	0.98
Communication	141	0.97
Overall	141	0.97

ordinal scale. The evaluation used 141 expert-generated questions reflecting anticipated user queries, each paired with a reference answer and supporting source materials.

To improve efficiency and reduce costs, we also used Ragas for LLM-based automated evaluation, applying an LLM-as-a-judge approach. The model scored chatbot responses using the rubric, human-written reference answers, and retrieved RAG content.

In the evaluation, we aimed to answer five research questions, and our key findings are following:

RQ 1. Does the Ask CSO retrieval process and bot configuration produce quality answers? Our results indicate that Ask CSO performed well across all three CCC dimensions (Correctness, Completeness, and Communication). The chatbot achieved a pass rate of 86% in Correctness, 85% in Completeness, and 90% in Communication, with an overall pass rate of 86%. These findings suggest that Ask CSO is a reliable tool for answering user questions effectively.

RQ 2. Are there any automated evaluation metrics that are strong proxies of human assessment of Ask CSO answers? To assess the effectiveness of the LLM-as-a-judge automated evaluation method, we measured Quadratic Weighted Kappa (QWK) between the two human raters (inter-human) and compared it to the QWK between each human rater and the LLM automated evaluator (human-to-LLM). Results showed no significant difference at the 95% confidence level between inter-human and human-to-LLM inter-rater reliability across all dimensions, except for Communication, where human-to-LLM agreement was significantly higher. This could mean that human reviewers had disparate interpretations of the Communication rubric and the LLM tended to produce ratings closer to the average human reviewer. Further research is required to confirm. These findings suggest that the LLM-as-a-judge method is a viable alternative to human evaluators for Ask NCES RAG chatbot evaluation.

RQ 3. How does the quality of Ask CSO answers compare with the quality of answers from GPT-4o? We compared Ask CSO's performance to GPT-4o with the automated evaluation approach. The evaluation results showed that Ask CSO achieved statistically significant improvements in pass rates across all dimensions. With an overall passing rate exceeding 89% across the CCC dimensions, Ask CSO outperformed GPT-4o by 8% to 12%.

RQ 4. Does adding examples and chain-of-thought to the rubric improve the LLM evaluator? We examined whether adding examples and chain-of-thought reasoning to rubrics improved the LLM evaluator's alignment with human ratings. Three rubric types were tested: a basic rubric ("No Examples"), one with example Q/A pairs ("Examples"), and another adding human reasoning ("Examples and Reasoning"). Using 141 test questions, Ragas evaluated responses with each rubric, and human-to-LLM IRR was measured via QWK. Results showed slightly higher correctness QWK as rubrics grew more detailed, while completeness QWK remained similar across rubrics and to inter-human IRR. Communication QWK was significantly higher for all LLM rubrics than inter-human IRR, suggesting the LLM applied criteria more consistently. Statistical tests confirmed these differences.

Additionally, IRR among the three LLM rubric types was extremely high (lowest QWK = 0.86), indicating all rubrics produced similar evaluations. While the LLM's higher consistency may stem from uniform criteria application, humans still offer nuanced judgment. The findings suggest detailed rubrics including a “few shot” approach ("Examples") and the Chain-of-Thought prompting ("Examples and Reasoning") marginally improve correctness alignment but may not be necessary for high IRR, as even basic rubrics ("No Examples") yielded strong agreement. Further research is needed to assess whether higher IRR reflects better evaluation quality or just stricter consistency.

RQ 5. How consistent is the LLM evaluator across successive iterations with identical parameters? To assess the LLM evaluator's (Ragas) consistency, the study ran 10 identical evaluations (using the "No Examples" rubric) on 141 test questions and measured pairwise QWK across iterations. The mean QWK for all dimensions exceeded 0.97, demonstrating near-perfect agreement and indicating minimal variability due to stochastic factors. This confirms the LLM evaluator produces highly consistent results under identical conditions. Default Ragas parameters were used (temperature = 0.01, top_p = 0.1), which are intended to produce deterministic results so this finding is expected.

Prior to introducing the LLM-as-a-judge evaluator, the study established a construct framework grounded in domain expertise. Subject-matter experts developed 141 evaluation questions spanning five topic areas, along with reference answers and their source links. Based on this framework, detailed rubrics were created and calibrated involving three independent human raters. This process provided content- and interpretive-validity evidence by anchoring evaluation criteria in expert judgment and establishing shared human interpretations of the constructs. The LLM evaluator was subsequently used to scale assessment under this predefined framework, supporting consistency while relying on human-defined constructs rather than model-internal notions of quality.

Nevertheless, as noted in prior work on LLM-as-a-judge methods, automated evaluators may exhibit highly consistent yet systematically biased scoring behavior, depending on factors such as rubric design, prompt framing, and domain context (Cheny et al., 2024; Li et al., 2025; Thakur et al., 2025). For this reason, LLM-based evaluation should not be used as a standalone substitute for human judgment. Although the observed stability supports its use for scalable monitoring and comparative analysis in low-stakes or exploratory settings, reliance on automated judgments alone may be inappropriate in high-stakes, highly subjective, or fairness-sensitive contexts without complementary human validation, periodic calibration, or audit procedures.

Insights

The evaluation results indicate that the proposed RAG-based framework can effectively retrieve and synthesize information from domain-specific reference materials, such as the NCES Statistical Stansdards Program materials. These findings suggest that similar architectures may be well suited for building reliable assistant systems to support users seeking structured, standards-based information in comparable domains.

Furthermore, this study highlights the broader potential of automating chatbot evaluation using an LLM-as-a-judge approach. The automated method demonstrated agreement levels comparable to those of human raters, suggesting that it can serve as a scalable and cost-effective alternative for ongoing evaluation. More broadly, this approach can be generalized to the assessment of other chatbot systems, helping to increase evaluation efficiency and reduce annotation costs in similar settings.

Limitations and future directions

While Ask CSO demonstrates strong performance in retrieving and synthesizing information, it is essential to recognize that the chatbot should be used as an assistant rather than a replacement for human expertise. The evaluation results indicate that in a few cases, neither Ask CSO nor GPT-4o yielded fully correct or complete answers, a limitation also observed in the evaluation of Ask NAEP (Zhang et al., 2024). These instances highlight the inherent limitations of Generative AI-driven chatbot systems, particularly in handling complex or nuanced queries that may require deeper contextual understanding or further clarification.

In addition, at the time of this study, GPT-4o and related LLM techniques had limited capability for processing statistical figures and numeric tables. As a result, the study design focused primarily on retrieval and synthesis of information from web-based and PDF text sources. Future work should explore advances in RAG approaches that more effectively support graphical and tabular data retrieval and reasoning.

Another direction for future research is to evaluate the proposed framework using multiple LLMs and newer model variants as they continue to evolve. This study relied on a single LLM, GPT-4o, due to platform and resource limitations; however, comparing performance across different models would provide insight into the robustness and generalizability of the findings. Given the rapid pace of development in LLM capabilities, future work should also examine how advances in model architectures and training data affect both chatbot performance and the reliability of LLM-based evaluators.

Finally, human oversight is critical in ensuring the accuracy and reliability of responses to NCES user queries. We suggest users leverage Ask NCES as an assistant in retrieving relevant information quickly while maintaining a verification process for correctness and completeness. Future developments should focus on refining the chatbot's capabilities while emphasizing its role as a support system rather than an autonomous decision-maker.

Appendix A

Technology of the Ask CSO Development

User interface design

The Ask CSO chatbot is one of multiple agents situated within the broader Ask NCES application. Users enter a landing page where they select the agent they would like to chat with. The user is then taken to the messaging window for that agent, which interacts exclusively with that agent's knowledge base. After submitting a question, the UI will display loading text to indicate the agent is preparing its response. The final response will include links to the web pages or PDFs that were used to provide an answer. Once a response has been generated, the user has the option to rate the response using a thumbs up/thumbs down button. Users can easily navigate back to the landing page by clicking the sign at the bottom left corner.

Frontend

The chatbot's front-end layer consists of a user interface (UI) built using the Flask framework and facilitates smooth and intuitive user interactions. This UI is designed to accommodate text-based inputs, with potential future enhancements allowing multimedia. After the user submits their query, the backend handles the querying and generation, and the responses are then processed into Markdown format for enhanced readability before being displayed in the user interface. This approach ensures high quality interactions and delivers responses in a clear, user-friendly format.

Relevance Check The system first utilizes NeMo Guardrails (Rebedea et al., 2023) to determine whether a question is relevant to the National Center for Education Statistics (NCES). Unlike traditional word-based filtering, this guardrail approach performs a self-check by comparing the input against a predefined set of irrelevant questions. The idea behind this method is to establish a safety constraint that mitigates vulnerabilities in LLMs. Compared to prompt engineering, the use of guardrails offers significant advantages in terms of reproducibility and reliability. Moreover, in addition to enabling more controlled dialogue management, this guardrail is programmable, which greatly facilitates the long-term maintenance and adaptability of the future LLMs.

1. **Database Search:** If the question passed the self-check of the guardrail, the system searches for an answer in our curated knowledge base.
2. **Summarization:** A generative AI agent summarizes the information found in the database and answers the question using the context retrieved.
3. **Web Search:** If additional information is needed, the system calls a web search agent to find answers from relevant reputable websites (e.g., government websites) on the internet. The agent then reads through the context of websites and tries to answer the question by filtering useful and relevant information.
4. **Combined Response:** The final generative AI agent combines the information from both the database and the web search to provide a comprehensive and summarized answer to the user's question.

Backend

The backend is engineered to handle requests efficiently while retrieving precise context from the RAG knowledge base. The setup filters user queries, accepting only relevant questions through the generative AI layer to obtain a final response. Leveraging AIR’s Azure OpenAI instance, the backend serves as the processing core for question-and-answer interactions, managing user requests and maintaining session persistence. Queries are run through our vector database to retrieve relevant documents, after which the query and document contents are passed to the text completions endpoint. Once the response is received, it is relayed to the frontend, ensuring a seamless user experience while optimizing the handling of data and AI interactions.

LLM We employ Azure OpenAI's generative AI models to manage various functions within our chatbot, including embedding data, conducting question-and-answer interactions, summarizing information, and facilitating web searches. At present, our chatbot primarily operates using the GPT-4o model to handle a range of tasks. The chatbot architecture is model agnostic, allowing for seamless integration of newly released models to enhance the chatbot's accuracy and fairness.

Knowledge Base and Data Storage. Our curated knowledge base includes publicly available information obtained from the NCES website and is maintained using LanceDB, an open-source vector database designed for managing, querying, and retrieving embeddings on large-scale datasets. The contents of our knowledge base come from the web pages and PDFs found under the *Statistical Standards Program* section of the NCES site (https://nces.ed.gov/statprog/).

Before importing the data into LanceDB, the collected texts from each webpage and PDF page were split into chunks using semantic chunking. These chunks maintain a sliding context window, with small amounts of preceding and consequent document text prepended and appended, respectively. Embeddings were generated for each chunk using OpenAI's *text-embedding-ada-002* model. The documents and their embeddings were then added to a LanceDB collection for fast retrieval.

We selected LanceDB as the primary database for storing embedded data. LanceDB offers a robust set of features that enable efficient data management and seamless access, allowing us to maintain the database with minimal effort. It natively supports advanced retrieval methods like hybrid search, as well as SQL style pre-filtering at query time.

Data security

In accordance with compliance standards and AIR AI use guidelines, all data is securely stored on the AIR server. Furthermore, the models used for generating embeddings and text completions are exclusively from AIR's Azure OpenAI resource, ensuring that sensitive operations are performed in a controlled, secure setting.

Appendix B

Prompt examples

1. Ragas with "No Examples" rubric–just rubric with no examples:

```
Correctness {
   "1": "Significant factual errors or misinformation",
   "2": "Some inaccuracies or lack of precision",
   "3": "Several minor inaccuracies, but generally correct",
   "4": "Generally accurate with 1-2 minor inaccuracies",
   "5": "Completely correct with no errors",
}
```

1. Ragas with "Examples" rubric–add 3 example(s) question/answer pairs to rubric, with question/answer pair and human evaluation score as examples:

{
 "1": "Significant factual errors or misinformation. ***Example 1:***

Question: Can I apply for more than one RUD with the same application?

Expected Answer: Multiple databases can be requested in one application, but please only request those databases that are directly related to your research questions/objectives. A research project description must be provided for each database. You would also need to provide specific justification for each database as to why you need access to the restricted-use version and why the publicly available data cannot be used to conduct your analysis.\n\nEach requested database must be listed in the online License application (Formal Request) and on each Affidavit of Nondisclosure.

Actual Answer: No, you cannot apply for more than one Restricted-Use Data (RUD) with the same application. Each application is specific to a single RUD license. If you need access to additional restricted-use data and you already have an active license, you should log in to your current existing license and submit an "add-data" request. If you need to designate more than seven staff members, you will have to apply for a second license

nd notify IES via email before doing so.

Correctness Score: 1

Example 2: … ,

"2": "Some inaccuracies or lack of precision…",

…

}

2. Ragas with "Examples and Reasoning" rubric–same as above, but humans explain their thought process for assigning the score:

{
 "1": "Significant factual errors or misinformation. Example 1:

Question: Can I apply for more than one RUD with the same application?

Expected Answer: Multiple databases can be requested in one application, but please only request those databases that are directly related to your research questions/objectives. A research project description must be provided for each database. You would also need to provide specific justification for each database as to why you need access to the restricted-use version and why the publicly available data cannot be used to conduct your analysis.\n\nEach requested database must be listed in the online License application (Formal Request) and on each Affidavit of Nondisclosure.

Actual Answer: No, you cannot apply for more than one Restricted-Use Data (RUD) with the same application. Each application is specific to a single RUD license. If you need access to additional restricted-use data and you already have an active license, you should log in to your current existing license and submit an "add-data" request. If you need to designate more than seven staff members, you will have to apply for a second license and notify IES via email before doing so.

Correctness Score: 1

Reasoning: The expected answer indicates that multiple databases can be requested in one application, but the actual answer from the bot is completely incorrect. It indicates that you cannot apply for multiple databases in a single application.

Example 2: … ,

"2": "Some inaccuracies or lack of precision…",

…

}

Acknowledgements
This work was conducted while Ting Zhang, Luke Patterson and Blue Webb were affiliated with AIR, which reviewed and approved the manuscript prior to submission. We would like to express our gratitude to Paul Bailey and Emmanuel Sikali for their contributions to the development of the Ask NCES chatbot. Paul played a key role in conceptualizing and developing the evaluation framework and rubric. Emmanuel provided project oversight and offered insightful feedback throughout both the development and evaluation phases.

Author contributions
Ting Zhang, Funding acquisition, Conceptualization, Methodology, Material and data preparation, Supervision, Writing—original draft, review and editing. Luke Patterson, Conceptualization, Methodology, Material and data preparation, Formal analysis, Writing—original draft. Blue Webb, Conceptualization, Methodology, Material and data preparation, Formal analysis, Writing—original draft. Zeyu Jin, Methodology, Formal analysis, Writing—original draft. Maggie Beiting-Parrish, Conceptualization, Methodology, Material and data preparation, Writing—review and editing. All authors read and approved the final manuscript.

Funding
U.S. Department of Education, ED-IES-12-D-0002/0004, 91990022C0053, and 91990023D0006/91990023F0350 with the American Institutes for Research.

Data availability
The evaluation data used and analyzed in this study are not publicly available due to privacy and confidentiality considerations. However, de-identified data may be made available from the corresponding author upon reasonable request and with appropriate permissions.

Declarations

Competing interests
The authors declare no competing interests.

Received: 17 October 2025 / Accepted: 4 February 2026
Published online: 12 March 2026

References
American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. American Educational Research Association.
Barbella, M., & Tortora, G. (2022). Rouge metric evaluation for text summarization techniques. Available at SSRN 4120317.
Cavnar, W. B., & Trenkle, J. M. (1994). N-gram-based text categorization. In *Proceedings of SDAIR-94, 3rd annual symposium on document analysis and information retrieval* (Vol. 161175, p. 14).
Chan, C., Chen, W., Su, Y., Yu, J., Liu, Z. (2023). ChatEval: Towards better LLM-based evaluators through multi-agent debate. Preprint. arXiv:2308.07201v1. https://arxiv.org/abs/2308.07201
Cheny, G. H., Cheny, S., Liu, Z., Jiang, F., & Wang, B. (2024). *Humans or LLMs as the judge? A study on judgement bias*. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing.
Cook, D. A., & Beckman, T. J. (2006). Current concepts in validity and reliability for psychometric instruments: Theory and application. *American Journal of Medicine*. https://doi.org/10.1016/j.amjmed.2005.10.036. (validity evidence sources).
Es, S., James, J., Espinosa-Anke, L., Schockaert, S. (2023). Ragas: Automated evaluation of retrieval augmented generation. Preprint. arXiv: 2309.15217. https://arxiv.org/abs/2309.15217
Fu, J., Ng, S., Jiang, Z., & Liu, P. (2023). GPTScore: Evaluate as you desire. Preprint. arXiv:2302.04166v2. https://arxiv.org/abs/2302.04166
Grice, P. (1989). *Studies in the way of words*. Harvard University Press.
Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., & Guo, J. (2024). *A Survey on LLM-as-a-Judge*. Preprint. arXiv: 2411.15594. https://arxiv.org/abs/2411.15594
Howcroft, D. M., & Rieser, V. (2021). What happens if you treat ordinal ratings as interval data? Human evaluations in NLP are even more under-powered than you think. In M.-F. Moens, X. Huang, L. Specia, & S. W. Yih (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 8932–8939). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.emnlp-main.703
Iskender, N., Polzehl, T., & Möller, S. (2021). Reliability of Human Evaluation for Text Summarization: Lessons Learned and Challenges Ahead. In A. Belz, S. Agarwal, Y. Graham, E. Reiter, & A. Shimorina (Eds.), *Proceedings of the Workshop on Human Evaluation of NLP Systems* (HumEval) (pp. 86–96). Association for Computational Linguistics. https://aclanthology.org/2021.humeval-1.10
Li, D., Jiang, B., Huang, L., Beigi, A., Zhao, C., Tan, Z., & Liu, H. (2024). *From Generation to Judgment: Opportunities and Challenges of LLM-as-a-Judge*. Preprint. arXiv: 2411.16594. https://arxiv.org/abs/2411.16594
Li, Q., Dou, S., Shao, K., Chen, C., & Hu, H. (2025). Evaluating Scoring Bias in LLM-as-a-Judge. *arXiv preprint* arXiv:2506.22316.
Li, H., et. al. (2024). LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods. Preprint. ArXiv: 2412.05579. https://arxiv.org/abs/2412.05579
Liang, T., He, Z., Jiao, H., Wang, X., Wang, Y., Wang, R., Yang, Y., Tu, Z., & Shi, S. (2023). Encouraging divergent thinking in large language models through multiagent debate. Preprint. arXiv:2305.19118. https://arxiv.org/abs/2305.19118
Lin, C. Y. (2004, July). Rouge: A package for automatic evaluation of summaries. In Text summarization branches out (pp. 74–81).
National Center for Education Statistics. (n.d.). *Statistical standards program*. U.S. Department of Education. https://nces.ed.gov/statprog/standards.asp

Ngyuen, M., Luo, L., Shiri, F., Phung, D., Li, Y., Vu, T., & Haffari, G. (2024). Direct evaluation of Chain-of-Thought in multi-hop reasoning with knowledge graphs. Preprint. arXiv: 2402.11199. https://arxiv.org/html/2402.11199v1

Papineni, K., Roukos, S., Ward, T., & Zhu, W. J. (2002, July). Bleu: a method for automatic evaluation of machine translation. In Proceedings of the 40th annual meeting of the Association for Computational Linguistics (pp. 311–318).

Rebedea, T., Dinu, R., Sreedhar, M., Parisien, C., & Cohen, J. (2023). NeMo Guardrails: A toolkit for controllable and safe LLM applications with programmable rails. Preprint. arXiv:2310.10501. https://arxiv.org/abs/2310.10501

Reiter, E. (2018). A structured review of the validity of BLEU. *Computational Linguistics, 44*(3), 393–401.

Schoch, S., Yang, D., & Ji, Y. (2020). "This is a Problem, Don't You Agree?" Framing and Bias in Human Evaluation for Natural Language Generation. In S. Agarwal, O. Dušek, S. Gehrmann, D. Gkatzia, I. Konstas, E. Van Miltenburg, & S. Santhanam (Eds.), Proceedings of the 1st Workshop on Evaluating NLG Evaluation (pp. 10–16). Association for Computational Linguistics. https://aclanthology.org/2020.evalnlgeval-1.2

Smith, E. M., Hsu, O., Qian, R., Roller, S., Boureau, Y.-L., & Weston, J. (2022). Human Evaluation of Conversations is an Open Problem: Comparing the sensitivity of various methods for evaluating dialogue agents. Preprint. arXiv:2201.04723.http://arxiv.org/abs/2201.0472

Szymanski, A., Ziems, N., Eicher-Miller, H. A., Li, T. J. J., Jiang, M., & Metoyer, R. A. (2025, March). Limitations of the LLM-as-a-judge approach for evaluating LLM outputs in expert knowledge tasks. In Proceedings of the 30th International Conference on Intelligent User Interfaces (pp. 952–966).

Thakur, A. S., Choudhary, K., Ramayapally, V., & Hupkes, D. (2025). Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-Judges. *arXiv*. https://doi.org/10.48550/arXiv.2406.12624

Vach, M., Gliem, M., Weiss, D., Ivan, V. L., Hauke, F., Boschenriedter, C., & Caspers, J. (2025). Evaluating Retrieval Augmented Generation-enhanced Large Language Models for Question Answering On German Neurovascular Guidelines. Clinical Neuroradiology, 1–9.

van der Lee, C., Gatt, A., van Miltenburg, E., & Krahmer, E. (2021). Human evaluation of automatically generated text: Current trends and best practice guidelines. *Computer Speech & Language, 67*, Article 101151. https://doi.org/10.1016/j.csl.2020.101151

Zhang, T., Patterson, L., Beiting-Parrish, M., Webb, B., et al. (2024). Ask NAEP: A generative AI assistant for querying assessment information. *Journal of Measurement and Evaluation in Education and Psychology, 15*(Special Issue), 378–394. https://doi.org/10.21031/epod .

Publisher's note